

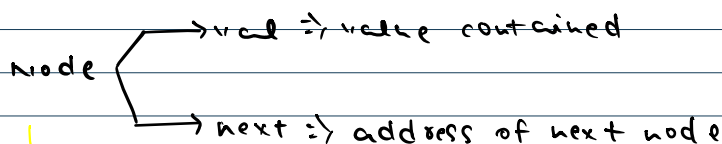
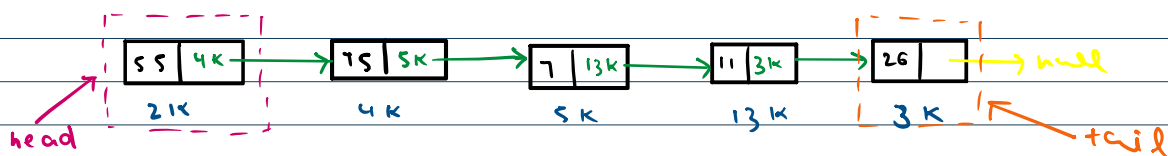
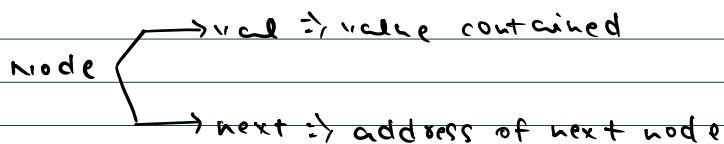
size of int

$$\begin{aligned} \text{arr}[5] &= \text{arr} + 4 \times 5 \\ &= 7K + 4 \times 5 \\ &= 7020 \end{aligned}$$

size fixed.
homogeneous

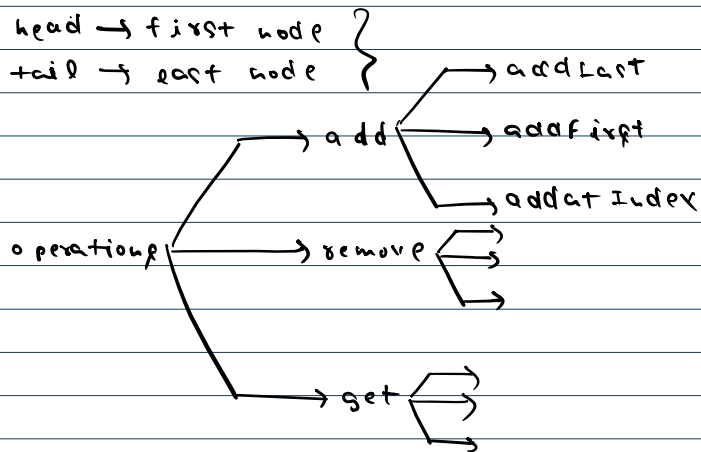
Linked List

It is a linear data structure where elements (called nodes) are connected through reference or pointer.



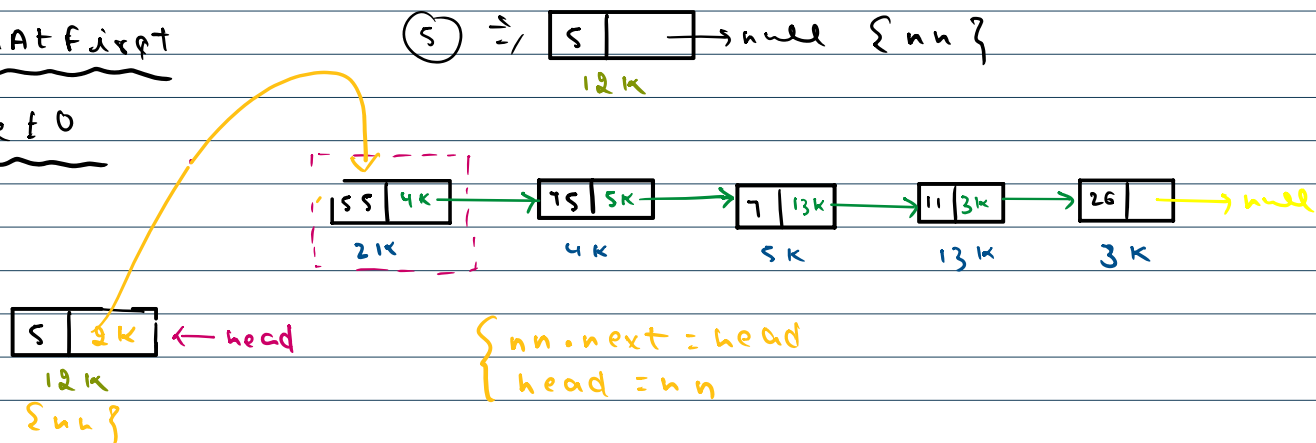
user-defined ⇒ class
data structure

```
class Node {
    int val;
    Node next;
}
```

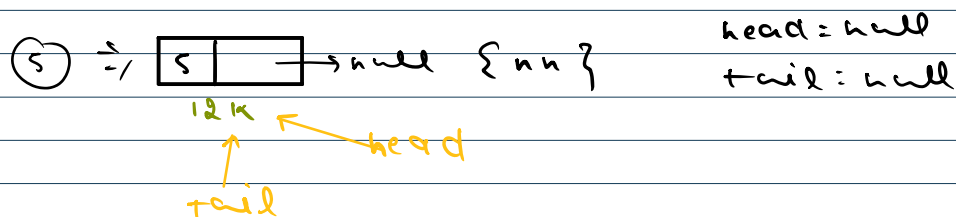


addAtFirst

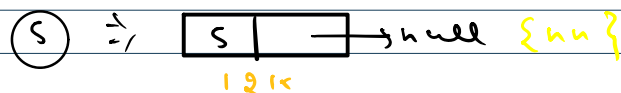
size $\neq 0$



size $= 0$

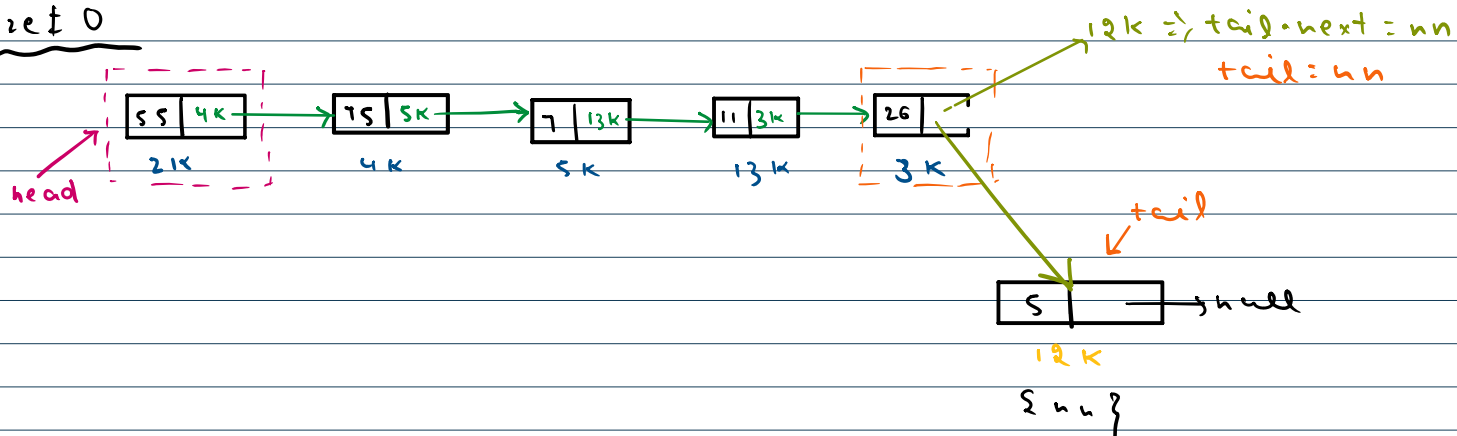


addAtLast



tail = 3k

size $\neq 0$

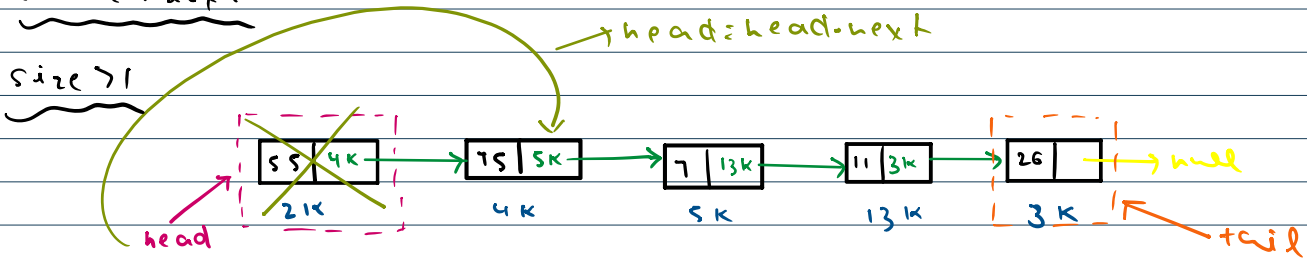


removeFirst

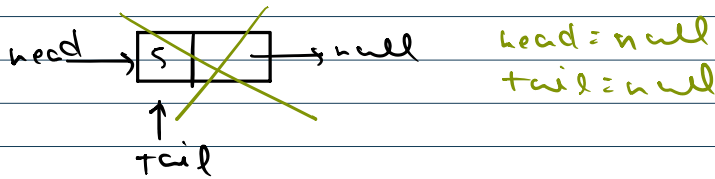


remove First

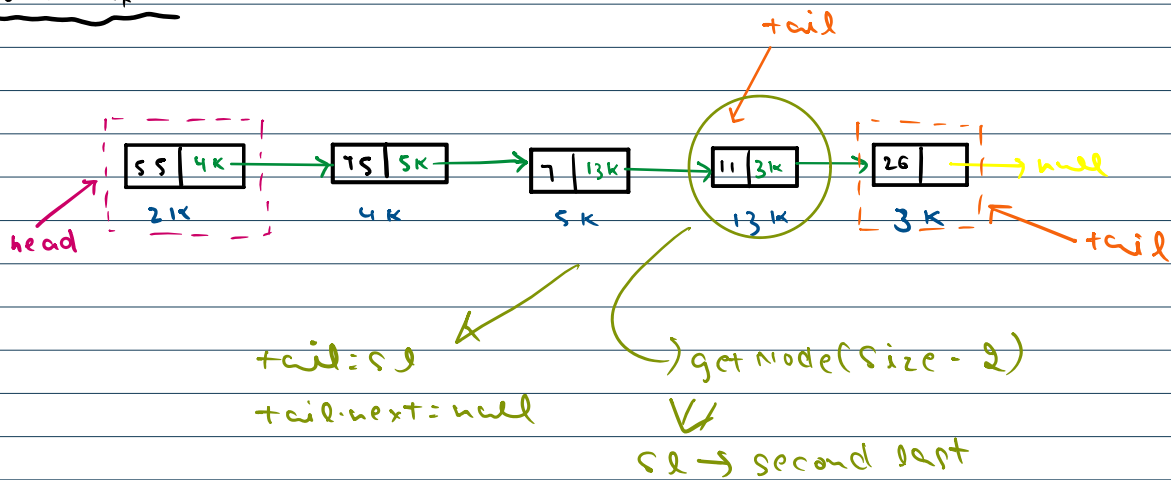
size > 1



size == 1



remove Last



addAtIndex

index = 2

(5) ⇒

5	
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 → null {nn}

